

Jorge Vazquez

765-772-8684 | jorgeaskur06@gmail.com | [Personal Webpage](#) | [in Jorge Vazquez](#) | [G JorgeAskur](#) | [Google Scholar](#)

EXPERIENCE

• CGVLab at Purdue University

Research Assistant

January 2025 - Currently

West Lafayette, Indiana

- Advisor: Bedrich Benes
- Working on procedural modeling in agricultural and biomedical scenarios. Creating a novel locomotion interface for VR systems.
- Created an interactive application and realistic terrain renderings in Vue for the TerraCover project.
- Developed an AI conversational agent for the VR-Bio-Talk project, based on a Large Language Model, allowing complex interaction via natural language.

• CGVLab at Purdue University

Visiting Scholar

August 2023 - December 2023

West Lafayette, Indiana

- Advisor: Voicu Popescu
- Developed a novel voice-based interface for Virtual Reality (VR), leveraging Large Language Models for complex and fast interaction in VR.
- Designed and conducted a user study with 22 participants to evaluate the voice-based interface, resulting in an improvement over conventional interfaces in various metrics.
- Tested complex virtual environments in VR and gathered user visibility error percentage data for an accepted conference publication.

• AirDX Technologies

Software Engineer Intern

February 2023 - July 2023

Monterrey, Mexico

- Implemented a simulated quadrotor with a LiDAR and Stereo Camera in Gazebo, which allowed the simulation of an inspection for damage detection on an A321 aircraft.
- Leveraged state-of-the-art Simultaneous Localization and Mapping (SLAM) algorithms in a UAV, achieving accurate movement in restricted environments.

EDUCATION

• Purdue University

Computer Science Ph.D. student at the Computer Graphics and Visualization Laboratory (CGVLab).

August 2024 - Currently

West Lafayette, Indiana

- Advisor: Bedrich Benes
- Teaching Assistant for Computer Architectures course in Fall 2024.
- Relevant Coursework: Algorithm Design, Analysis and Implementation, Data Communication, Computer Networks, Distributed Systems, and Motion Planning.

• Tecnologico de Monterrey

B.S. in Robotics and Digital Systems Engineering

August 2020 - June 2024

Monterrey, Mexico

- GPA: 4.00/4.00
- Academic Merit Scholarship
- Student groups: VexU robotics ITESM2, VantTec autonomous vehicles group
- Relevant Coursework: Data Structures and Algorithms, Computational Architectures, Internet of Things, Object-Oriented Programming, Secure Networks, Robotics.

PROJECTS

• Development of an Unmanned Underwater Vehicle in a student group

Tools: ROS, Python, OpenCV, Docker, Gazebo

August 2022 - June 2024



- Built a 3D real-time position estimation system that leverages neural networks with YOLOv7 (You Only Look Once).
- Coordinated a team of seven people to design, document, and implement a state machine for a simulated Unmanned Underwater Vehicle (UUV).

• Puzzlebot Autonomous Challenge

Tools: ROS, Python, Jetson Nano, Linux, Docker

March 2024 - June 2024



- Coded various motion planning algorithms (Bug 0, Bug 1 & Bug 2), achieving obstacle avoidance using LiDAR data.
- Applied dead reckoning localization with an Extended Kalman Filter using camera data and known landmarks, allowing accurate localization in a course.
- Implemented SLAM Toolbox and NavStack, allowing the robot to use effective path planning algorithms like A* to navigate an obstacle course.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

-
- [C] JAV Fernandez, JJ Lee, SAS Vacca, AJ Magana, R Pesa, B Benes, V Popescu. (2025). **Hands-Free VR**. Accepted for publication in the *20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - HUCAPP*.
- [C] V Popescu, E Sacks, Z Zhang, JAV Fernandez (2025). **Complex Virtual Environments on All-In-One VR Headsets Through Continuous From-Segment Visibility Computation**. Accepted for publication in *IEEE VR 2025*.

SKILLS

-
- **Programming Languages:** Proficient (Python, C, C++, C#), Familiar (JavaScript, x86 Assembly, ARM Assembly)
 - **Frameworks and tools:** SQL, Firebase, Pandas, PyTorch, TensorFlow, OpenCV, NumPy, Git, Docker, HTML, CSS, Express, Flask, ROS, Unity, VR development, Embedded Systems, Communication Protocols (SPI, Serial, CAN, I2C)